

Solution

GRAVITY-02

Class 09 - Science

1. According to universal law of gravitation, the gravitational force of attraction between any two objects of mass is proportional to the product of the masses and inversely proportional to the square of the distance between them. Hence if the distance is reduced to half, then the gravitational force becomes four times larger than the previous value.

2. According to universal law of gravitation, two objects attract each other with equal force, but in opposite direction.

The earth and the moon are attracted to each other by same gravitational force because for both of them formula to calculate force of attraction is the same

$$F = G \frac{M_e \times M_m}{d^2} \quad d \text{ is also same for both.}$$

3. Given the weight of an object of the moon is one sixth weight of an object of the Earth. Also Weight = Mass x Acceleration

$$\text{Value of gravity on earth} = 9.8 \text{ m/s}^2$$

$$\text{value of gravity on moon} = 1/6^{\text{th}} \text{ of earth} = 9.8/6 = 1.63 \text{ m/s}^2$$

$$\text{weight of object on moon} = m \times 1.63 = 10 \times 1.63 = 16.3 \text{ N}$$

$$\text{weight of object on earth} = m \times 9.8 = 10 \times 9.8 = 98 \text{ N}$$

4. This is due to the reason that the force of gravitational attraction attracts the bodies towards each other but attraction between the bodies is very small when mass is accounted. On the other hand due to the large mass of the earth, the force of gravitational attraction between the earth and the bodies is large.

5. For the first stone, we are given,

$$u = 0 \text{ ms}^{-1}, h = 49 \text{ m}$$

Using the relation, $S = ut + \frac{1}{2}at^2$ we have

$$49 = 0 \times t + \frac{1}{2} \times 9.8 \times t^2 \Rightarrow t^2 = \frac{98}{9.8} = 10 \Rightarrow t = \sqrt{10} = 3.16 \text{ s}$$

So, the first stone would take 3.16 s to reach the ground.

For the motion of the second stone: The second stone was thrown one second after the first stone. But, both the stones reach the ground at the same time. Thus, the time taken by the second stone to reach the ground is one second less than that taken by the first stone. So,

$$t_2 = t - 1 = (3.16 - 1) \text{ s} = 2.16 \text{ s}$$

Hence for the second stone, we have

$$U = ? \text{ ms}^{-1}, g = 9.8 \text{ ms}^{-2}, h = 49, t = 2.16 \text{ s}$$

Using $S = ut + \frac{1}{2}at^2$ we have

$$49 = u \times 2.16 + \frac{1}{2} \times 9.8 \times (2.16)^2 \Rightarrow 49 - 22.86 = 2.16 u$$

$$\text{or } 26.14 = 2.16 u \Rightarrow \frac{26.14}{2.16} = 12.10 \text{ ms}^{-1}$$

Thus, the second stone was thrown downward with a speed of 12.1 ms^{-1} .

6. Given that mass of the body $m = 1 \text{ Kg}$

$$\text{Mass of the Earth } M = 6 \times 10^{24} \text{ Kg}$$

$$\text{Radius of the Earth } R = 6.4 \times 10^6 \text{ m}$$

Now magnitude of the gravitational force (F) between the Earth and the body can be given as,

$$\begin{aligned} F &= G \frac{Mm}{d^2} \\ &= \frac{6.6 \times 10^{-11} \times 6 \times 10^{24} \times 1}{(6.4 \times 10^6)^2} \\ &= \frac{6.7 \times 6 \times 10}{(6.4 \times 6.4)} \\ &= 9.81 \text{ N} \end{aligned}$$

7. Let the distance between the top of the window and the roof be S .

a) For the journey across the window i.e., from B to C.

$$\text{Let, Velocity at B} = u \text{ ms}^{-1}$$

$$\text{Distance travelled, } S = 2 \text{ m}$$

$$\text{Time taken, } t = 0.1 \text{ s}$$

$$\text{Acceleration, } a = g = 9.8 \text{ ms}^{-2}$$

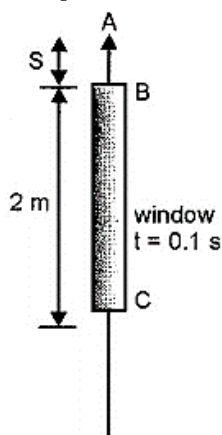
Using the equation $S = ut + \frac{1}{2}at^2$ we have

$$2 = u \times 0.1 + \frac{1}{2} \times 9.8 \times (0.1)^2 \Rightarrow u = 19.51 \text{ ms}^{-1}$$

Therefore the velocity of the stone at the top of the window is

19.51 ms^{-1} . Now for the free fall of the stone from the rooftop to the top of the window i.e.,

From point A to B we have



Initial velocity $u = 0 \text{ ms}^{-1}$

Final velocity $v = 19.51 \text{ ms}^{-1}$

Distance covered $S = ?$

Acceleration due to gravity $g = 9.8 \text{ ms}^{-2}$

Using $v^2 - u^2 = 2aS$, we have

$$(19.51)^2 - 0^2 = 2 \times 9.8 \times S \Rightarrow \frac{(19.51)^2}{19.6} = 19.42 \text{ m}$$

8. A greater surface area offers greater resistance and buoyancy same is true in the case of a sheet of paper that has larger surface area as compared to paper crumpled into a ball. The paper which is crumpled has higher density, hence its resistance to its motion through the air decreases. So sheet of paper falls slower.

9. Force of attraction between two objects is given by

$$F = \frac{Gm_1m_2}{r^2}$$

i. If the mass of one object is doubled then the force will be doubled.

ii. If the distance between the objects is doubled then the force becomes one-fourth and if the distance between the objects is tripled then the force becomes one-ninth.

iii. If masses of both objects are doubled then the force becomes four times.

10. The acceleration produced in a body due to force of gravity is called acceleration due to gravity. It is denoted by g . The value of acceleration due to gravity is taken as 9.8 ms^{-2} at 45° latitude, at the sea level.

11. i. The SI unit of thrust in newton (N). The SI unit of pressure in Nm^{-2} or pascal (Pa).

ii. We exert more pressure on the ground when we stand on one foot than both feet, as the area of one foot is half than that of two feet as $p \propto \frac{1}{A}$.

12. We know that, Acceleration due to gravity, $g = \frac{GM}{R^2}$ or $M = \frac{gR^2}{G}$

$\Rightarrow$ We know that, Average density of the earth, $D = \frac{\text{Mass}}{\text{Volume}} = \frac{gR^2}{G \times V_e}$ (where V_e is the volume of the Earth)

$$\text{or } D = \frac{gR^2}{G \times \frac{4}{3}\pi R^3} = \frac{3g}{4\pi GR}$$

13. Differences between acceleration due to (g) and Universal gas constant (G) are given below:

Acceleration due to gravity (g)	Universal gravitational constant (G)
1. Acceleration due to gravity is the acceleration acquired by a body due to the earth's gravitational pull on it.	1. The gravitational constant is numerically equal to the force of attraction between two masses of 1 kg that are separated by a distance of 1 m.
2. Acceleration due to gravity (g) is a vector quantity.	2. Universal gravitational constant (G) is a scalar quantity.
3. It is different in different places on the surface of the earth. Its value also varies from one celestial body to	3. The ' G ' is a universal constant, i.e., its value is the same (i.e. $6.7 \times$

another.	$10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$) everywhere in the universe.
4. Its value depends upon the size and nature of the bodies.	4. Its value does not depends upon the size and nature of the bodies.

14. The universal law of gravitation is important because it tells us about

- i) the force that binds us to earth.
- ii) this law describes the motion of planets around the sun.
- iii) this law justifies the tide formation on earth due to moon and sun.
- iv) this law gives reason for movement of moon around earth.
- v) Tides formed by rising and falling of water level in the ocean due to the gravitational force exerted by both Sun and the moon of the Earth.

15. The bag of cotton is heavier since volume of cotton bag is greater than iron bar, so the upthrust is larger in case of cotton. so the weighing machine indicates a smaller mass for cotton bag than its actual mass.

16. We know that, Acceleration due to gravity, $g = \frac{GM}{R^2}$, M= Mass of earth, R= radius of earth and G= universal gravitational constant.

Since, Acceleration due to gravity is independent of the mass of the falling body. Therefore, This hypothesis is not correct. The two bricks like a single body, fall with the same speed to reach the ground at the same time in case of free fall.

17. Given, H = 1000 m, v = 0, g = 9.8 m/s²

i. From the third equation of motion, $v^2 = u^2 + 2gH \Rightarrow 0 = u^2 - 2 \times 9.8 \times 1000$

$$u = \sqrt{2 \times 9.8 \times 1000} = \sqrt{19600} = 140 \text{ m/s}$$

ii. From the first equation of motion, $v = u + gt \Rightarrow 0 = 140 - 9.8 t$

$$\Rightarrow t = \frac{140}{9.8} = 14.28 \text{ s}$$

18. i. Given, initial velocity, u = 49 m/s

At the maximum height velocity becomes zero.

$\therefore$ Final velocity, v = 0

From the third equation of upward motion,

$$v^2 = u^2 - 2gh$$

$$0 = (49)^2 - 2 \times 9.8 \times h$$

$$\therefore h = \frac{49^2}{2 \times 9.8} = 122.5 \text{ m}$$

Maximum height attained = 122.5 m

ii. Time is taken by the ball to reach the maximum height.

From the first equation of motion, $v = u - gt$

$$\text{or } 0 = 49 - 9.8 \times t$$

$$t = \frac{49}{9.8} = 5 \text{ s}$$

For the motion against gravity, the time of descent is the same as the time of ascent. So, time taken by the ball to fall from maximum height is 5 s.

$\therefore$ Total time taken by the ball to return to the surface of the earth = 5 + 5 = 10 s

19. Initial velocity of stone (u) = 40 m/s

at maximum height stone will be at rest so v = 0

$$v = u + gt$$

$$0 = 40 + (-10) \times t$$

$$10t = 40$$

$$t = 40/10 = 4 \text{ s}$$

distance covered /maximum height

$$h = ut + \frac{1}{2}gt^2$$

$$= 40 \times 4 + \frac{1}{2} \times (-10) \times 4 \times 4$$

$$= 160 - 80 = 80 \text{ m}$$

net displacement of stone = 0 (thrown upwards then falls back to same place)

Total distance covered by the stone = 80 + 80 = 160 m

20. Suppose both the stones will meet after t seconds and h be the height from the ground. Given height of the tower is 100m

Consider the stone falls from the top of the tower. So distance covered by the stone at time t can be calculated according to

equation of motion

$$h = ut + \frac{1}{2}gt^2$$

$$= 0 \times t + \frac{1}{2} \times 10 \times t^2$$

$$= 5t^2$$

$$h = ut + \frac{1}{2}gt^2$$

$$= 25 \times t + \frac{1}{2} \times (-10) \times t^2$$

$$= 25t - 5t^2$$

$$h + h' = 100 \text{ m}$$

$$5t^2 + 25t - 5t^2 = 100$$

$$25t = 100$$

$$t = 4 \text{ s}$$

$$h = 5t^2 = 5 \times 4 \times 4 = 80 \text{ m}$$

Therefore, the two stones will meet after 4 seconds when the falling stone would have covered a height of 80 m.

21. Distance travelled, $s = 10 \text{ m}$

Final velocity, $v = 0 \text{ ms}^{-1}$

Acceleration due to gravity, $g = 9.8 \text{ ms}^{-2}$

Acceleration of the object, $a = -9.8 \text{ ms}^{-2}$ (upward motion)

Now, from 3rd law of motion,

$$\text{i. } v^2 = u^2 + 2as$$

$$0 = u^2 + 2 \times (-9.8 \text{ ms}^{-2}) \times 10 \text{ m}$$

$$-u^2 = -2 \times 9.8 \times 10 \text{ m}^2\text{s}^{-2}$$

$$u = \sqrt{196} \text{ ms}^{-1}$$

$$u = 14 \text{ ms}^{-1}$$

Thus,

$$\text{Initial velocity, } u = 14 \text{ ms}^{-1}$$

$$\text{ii. } v = u + at$$

$$0 = 14 \text{ ms}^{-1} - 9.8 \text{ ms}^{-2} \times t$$

$$t = 1.43 \text{ s.}$$

Thus,

Time is taken $t = 1.43 \text{ s}$.

22. Gravity is directly related to mass. The more mass an object has, the more gravitational pull it has.

Now, the moon is significantly smaller than the earth (in fact, it is about the size of the earth's core). Gravity is dependent on the size of the object. Your weight on the moon is 1/6 of that on earth because the moon has 1/6 the mass of the earth.

Since we know

$$W = m \times g$$

Mass of object remains the same whether on earth or moon but the value of acceleration on moon is $1/6^{th}$ of the value of acceleration on earth. Because of this weight of an object on moon is $1/6^{th}$ its weight on the earth.

23. The acceleration due to gravity on the moon is one-sixth the acceleration due to gravity on the earth. The radius of the moon is

$1.74 \times 10^6 \text{ m}$ and the value of the gravitational constant is $6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$

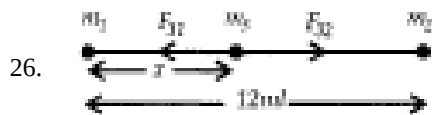
Therefore the acceleration due to gravity on the surface of the moon is

$$g_m = \frac{g_e}{6} = \frac{9.8}{6} = 1.63 \text{ ms}^{-2}. \text{ Using } g = \frac{GM}{R^2} \text{ we have}$$

$$M_m = \frac{g_m R_m^2}{G} = \frac{1.63 \times (1.74 \times 10^6)^2}{6.67 \times 10^{-11}} = 7.39 \times 10^{22} \text{ kg}$$

24. The gravitational force is the source of centripetal force that a planet requires to revolve around the sun. This force depends on the masses of the Sun and the planet and on distance between them.

25. Earth is a huge ball spinning about its axis. Due to the spinning, every object on the surface of the earth experiences a centrifugal force. This centrifugal force tries to throw the object outward away from the centre of the earth. However, the objects do not fly off the surface of the earth, because the force of gravity pulls the objects towards the centre of the earth, thereby balancing the centrifugal force. However, if the force of gravity somehow disappears, then centrifugal force will throw the bodies on the surface of the earth into space.



Given, Mass of one body, $m_1 = 3\text{kg}$; and Mass of other body, $m_2 = 12\text{kg}$.

Let the mass of third body, $m_3 = 0.5\text{ kg}$ be placed at a distance of 'x'm from m_1 , as shown in figure.

Then force acting on m_3 due to m_1 , is equal and opposite to the force acting on m_3 due to m_2 , i.e the force on third body is zero;

$$\therefore F_{31} = F_{32}$$

$$\frac{Gm_1m_3}{x^2} = \frac{Gm_3m_2}{(12-x)^2}$$

Put the given values, we have,

$$\frac{3}{x^2} = \frac{12}{(12-x)^2}$$

$$\Rightarrow \left(\frac{12-x}{x}\right)^2 = \frac{12}{3} = 4 \Rightarrow \frac{12-x}{x} = 2$$

$$\Rightarrow 12 - x = 2x \Rightarrow 12 = 3x$$

$$\Rightarrow x = 4\text{ m}$$

The position of required point is at a distance of 4 m from mass of 3 kg and

The position of required point is at distance of $(12\text{m} - 4\text{m})=8\text{m}$ from mass of 12 kg.

27. Let P and Q be the two bodies,

Let, the mass of the body P = m_1 and the mass of body Q = m_2

Let the mass of earth = M_e .

According to the universal law of gravitation, the force of attraction between the earth and the body P is given by,

$$F_P = \frac{G \times M_e \times m_1}{R^2} \dots\dots\dots(1)$$

Where R is the distance of the body from the centre of the earth.

Similarly, the force of attraction between the earth and the body Q is given by

$$F_Q = \frac{G \times M_e \times m_2}{R^2} \dots\dots\dots(2)$$

Since, the two forces, i.e., F_P and F_Q are equal, thus from (1) and (2),

$$\frac{G \times M_e \times m_1}{R^2} = \frac{G \times M_e \times m_2}{R^2}$$

$$\Rightarrow m_1 = m_2 \text{ (Hence proved)}$$

Therefore, if the earth attracts two bodies placed at the same distance from the centre of the earth with the same force, then their masses are equal.

28. Let M_E be the mass of the earth and m be the mass of the object on its surface. It R is the radius of the earth, then according to the universal law of gravitation, the gravitational force F acting between the earth and the object is given by the relation

$$F = G \frac{Mm}{r^2}$$

29. For a body thrown vertically upwards,

$$\text{Initial velocity} = u \text{ ms}^{-1}$$

$$\text{Final velocity} = v = 0 \text{ ms}^{-1}$$

Let t_1 be the time of ascent. Now for the vertically upward motion, the equation of motion is

$$v = u + at, \text{ which can be written as } v = u - gt$$

$$0 = u - gt_1 \Rightarrow t_1 = \frac{u}{g} \dots\dots (i)$$

For the return journey, when the body falls vertically downwards. Let the time of descent be t_2 . Also the equation of motion for the downward motion is

$$v = u + gt, \text{ but } u = 0, \text{ therefore}$$

$$v = 0 + gt_2 \Rightarrow t_2 = \frac{v}{g} \dots\dots (ii)$$

Since, $v = u$. Thus from (i) and (ii) we find that $t_1 = t_2$ i.e., the time of ascent is equal to the time of descent.

30. All objects fall on ground with constant acceleration, acceleration due to gravity (in the absence of air resistance). It is constant and does not depend upon the mass of an object. Hence, heavy objects do not fall than lighter objects.

31. For an object immersed in water two forces act on it

Gravitational force which tends to pull the object in downward direction

buoyance force which pushes the object in the upward direction. Since plastic has density very less as compared to water i.e.

weight of plastic is less than the buoyant force experienced by it therefore a block of plastic released under water come up to the surface of water/floats.

32. Let acceleration due to gravity at the equator (g_e). Let acceleration due to gravity at the pole = g_p

$$\text{Now, } g = \frac{GM}{R^2} \text{ Therefore we have } \frac{g_p}{g_e} = \frac{GM}{R_p^2} \times \frac{R_e^2}{GM} \times \frac{R_e^2}{R_p^2} = \frac{(6378)^2}{(6357)^2} = 1.0066$$

$\therefore$ Percentage change in the acceleration due to gravity at the poles

$$\frac{g_p - g_e}{g_e} \times 100 = \frac{1.0066g_e - g_e}{g_e} \times 100 = 0.66\%$$

33. Initial velocity of stone (u) = 0

Final velocity of stone (v) = ?

Height attained (S) = 19.6 m

Acceleration due to gravity (g) = + 9.8 ms⁻²

We know : $v^2 = u^2 + 2 gS$

$$v^2 = (0)^2 + 2 \times 9.8 \times 19.6$$

$$v^2 = 19.6 \times 19.6$$

$$v = \sqrt{19.6 \times 19.6}$$

$$v = 19.6 \text{ ms}^{-1}$$

34. Initial velocity, $u = 15 \text{ m/s}$

Final velocity, $v = 0$

From first equation of motion,

$$v = u + at$$

$$\Rightarrow 0 = 15 + at$$

$$\Rightarrow t = \frac{-15}{a} \text{ sec}$$

$\therefore$ time taken to reach the highest point = $\frac{-15}{a} \text{ sec}$

Time taken to reach downward = $\frac{-15}{a} \text{ s}$

$\therefore$ Time taken for the entire journey = $2 \left(\frac{-15}{a} \right)$

$$\Rightarrow 2 \left(\frac{-15}{a} \right) = 7.5 \text{ s}$$

Therefore, $a = -4 \text{ m/s}^2$

Therefore, the acceleration due to gravity on the planet = -4 m/s^2

35. For a ball thrown vertically upwards,

Initial velocity = $u \text{ ms}^{-1}$

Final velocity = $v = 0 \text{ ms}^{-1}$

For the vertically upward motion, the equation of motion is

$v = u + at$ which can be written as $v = u - gt$

$$0 = u - gt \Rightarrow t = \frac{u}{g} \dots\dots (i)$$

For the return journey, when the body falls vertically downwards, the equation of motion is

$v = u + gt$, but $u = 0$ therefore

$$v = 0 + gt \Rightarrow t = \frac{v}{g} \dots\dots (ii)$$

from (i) and (ii) we find that $v = u$. Thus a body returns to its point of projection with the same velocity with which it was thrown.

36. Initial velocity, $u = 0$

Let A fall for ' t ' seconds

acceleration, $a = -9.8 \text{ m/s}^2$

For Object A, Distance travelled = S_1

$$S_1 = ut + \frac{1}{2}gt^2$$

$$\Rightarrow S_1 = 0 - \frac{1}{2} \times 9.8t^2$$

$$\Rightarrow S_1 = -\frac{1}{2} \times 9.8t^2 \dots(i)$$

For object B, time = $(t - 1)\text{s}$

Distance travelled = S_2

$$S_2 = ut + \frac{1}{2}gt^2$$

$$\Rightarrow S_2 = -\frac{1}{2}9.8(t - 1)^2 \dots(ii)$$

Subtracting equation (ii) from (i)

$$S_1 - S_2 = -10 \text{ m}$$

$$\Rightarrow -10 = \frac{1}{2} \times 9.8 \left[(t-1)^2 - t^2 \right]$$

$$\Rightarrow -10 = 4.9 [t^2 - 2t + 1 - t^2]$$

$$\Rightarrow -10 = 4.9[-2t + 1]$$

$$\Rightarrow -10 = -9.8t + 4.9$$

$$\Rightarrow -9.8t = -14.9$$

$$\Rightarrow t = 1.5 \text{ sec}$$

The objects will be 10 m apart 1.52 s after A is dropped.

37. i. Initial velocity of the ball (u) = 49 ms^{-1}

Final velocity of the ball (v) = 0

Acceleration due to gravity (g) = -9.8 ms^{-2}

[In upward direction g is -ve]

Height attained by the ball (S) = ?

Time for rising up (t) = ?

We know : $v^2 - u^2 = 2gS$

$$(0)^2 - (49)^2 = 2 \times (-9.8) \times S$$

$$S = \frac{-49 \times 49}{-2 \times 9.8}$$

$$S = 122.5 \text{ m}$$

We know $v = u + gt$

$$0 = 49 - 9.8 \times t$$

$$\Rightarrow t = \frac{49}{9.8}$$

$$\Rightarrow t = 5 \text{ s}$$

ii. Now, time for upward journey of the ball

= the time for downward journey of the ball.

$\therefore$ Total time is taken by the ball to return to the surface of earth = $2 \times t = 2 \times 5 = 10 \text{ s}$

38. Universal law of gravitation: The universal law of gravitation states that every object in the universe attracts every other object with a force called the gravitational force. The force acting between two objects is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres.

39. i. Upthrust = loss in weight = $100 - 88 = 12 \text{ gf}$

ii. Weight of water displaced = upthrust = 12 gf

Since density of water is 1 gcm^{-3} therefore volume of water displaced = 12 cm^3

But the solid displaces water equal to its own volume. Therefore the volume of solid = 12 cm^3

$$\text{iii. RD of solid} = \frac{\text{weight in air}}{\text{weight in air} - \text{weight in water}} = \frac{100}{100 - 88} = 8.33$$

40. When all other variables remain constant, the force of gravitation is inversely proportional to the square of distance between the two objects.

$$F \propto \frac{1}{r^2} \Rightarrow F \propto \frac{1}{(1/2)^2} = 4$$

$\therefore$ the force of gravitation increases 4 times.

41. The bag of sugar will weigh more at Antarctica i.e. the poles.

Reason: The polar radius of the earth is less than the equatorial radius of the earth. The acceleration due to gravity is inversely proportional to the square of the radius of the earth. Hence, its value at the poles is greater than its value at the equator. Thus a body weighs more at the poles than at the equator. Since weight is the product of mass and acceleration due to gravity, therefore the bag of sugar will weigh more at Antarctica.

42. Mass of object (m) = 10 kg

Acceleration due to gravity on earth (g_e) = 9.81 ms^{-2}

Acceleration due to gravity on moon (g_m)

$$= \frac{9.81}{6} \text{ ms}^{-2}$$

Weight of the object on the earth = mg_e

$$= 10 \times 9.81 = 98.1 \text{ N}$$

Weight of the object on the moon = mg_m

$$= \frac{10 \times 9.81}{6} = 16.35 \text{ N}$$

43. Mass of the moon (M) = $7.4 \times 10^{22} \text{ kg}$

Radius of the moon (R) = $1.74 \times 10^6 \text{ m}$

Gravitational constant (G) = $6.7 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$

$\therefore$ Acceleration due to gravity = $\frac{GM}{R^2}$

On moon (g_m)

$$g_m = \frac{6.7 \times 10^{-11} \times 7.4 \times 10^{22}}{(1.74 \times 10^6)^2}$$

$$= \frac{6.7 \times 7.4}{1.74 \times 1.74} \times \frac{10^{-11+22}}{10^{12}}$$

$$g_m = 1.63 \text{ ms}^{-2}$$

$$\frac{g_m}{g_e} = \frac{1.63}{9.81} = \frac{1}{6} \text{ approx.}$$

$$g_m = \frac{1}{6} \text{ approx.}$$

44. Here, initial velocity of stone, $u = 0$

Time is taken by stone to reach the ground, $t = 4 \text{ s}$

Acceleration, $a = g = 9.8 \text{ m/s}^2$

Let, the height of the building, $h = ?$

Using the equation of motion, $S = ut + \frac{1}{2}at^2$.

$$h = ut + \frac{1}{2}gt^2 = 0 + \frac{1}{2}gt^2$$

$$h = \frac{1}{2} \times 9.8 \text{ m/s}^2 \times (4\text{s})^2$$

$$= \frac{1}{2} \times 9.8 \times 16 \text{ m} = 78.4 \text{ m}$$

$$\Rightarrow h = 78.4 \text{ m.}$$

Therefore, height of the building, $h = 78.4 \text{ m.}$

45. Distance travelled, $h = 13.07 \text{ m}$

Final velocity $v = 0$

Acceleration due to gravity $g = -9.8 \text{ m/s}^2$ (upward motion)

$$\text{i. } v^2 = u^2 + 2gh \Rightarrow 0 = u^2 + 2 \times (-9.8) \times 13.07$$

$$u^2 = 256 \Rightarrow u = 16 \text{ m/s}$$

$$\text{ii. } v = u + at \Rightarrow 0 = 16 - 9.8 \times t \Rightarrow t = 1.63 \text{ s.}$$

46. It is difficult to hold a school bag having a thin strap because the pressure on the shoulders is quite large. This is because the pressure is inversely proportional to the surface area on which the force acts. The smaller is the surface area the larger would be the pressure on the surface. In the case of a thin strap, the contact surface area is very small. Hence pressure exerted on the shoulder is very large.

47. The acceleration due to gravity is more at the poles than at the equator. When the initial velocities and the distance travelled are the same, time taken for a body is smaller if the acceleration due to gravity is more. Hence when dropped from the same height a body reaches the ground quicker at poles than at the equator.

48. We know, that $G = 6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$

Mass of the earth, $M_e = 6 \times 10^{24} \text{ kg}$

And Radius of the earth, $R_e = 6.4 \times 10^6 \text{ m}$

We know that, acceleration due to gravity, $g = \frac{G \times M_e}{R_e^2}$.

$$\therefore g = \frac{6.67 \times 10^{-11} \times 6 \times 10^{24}}{(6.4 \times 10^6)^2} \text{ m/s}^2$$

$$\Rightarrow g = \frac{6.67 \times 6 \times 10}{6.4 \times 6.4} \text{ m/s}^2 = 9.8 \text{ m/s}^2.$$

Therefore, the value of acceleration due to gravity, g on the surface of earth = 9.8 m/s^2 .

49. Case (i)

Initial velocity of the ball (u) = 0

The final velocity of the ball (v) = ?

Distance covered (S) = 20 m

Acceleration due to gravity (g) = 10 ms^{-2}

Using $v^2 - u^2 = 2aS$ we have

$$v^2 - 0^2 = 2 \times 10 \times 20 = 400 \text{ or } v^2 = 400 \Rightarrow v = 20 \text{ ms}^{-1}$$

Case (ii)

Initial velocity of the ball (u) = 0

Final velocity of the ball (v) = ?

Distance covered (S) = 40 m

Acceleration due to gravity (g) = 10 ms^{-2}

Using, $v^2 - u^2 = 2aS$ we have

$$v^2 - 0^2 = 2 \times 10 \times 40 = 800 = 2 \times 400 \text{ or } v^2 = 2 \times 400 \Rightarrow v = 20\sqrt{2} \text{ ms}^{-1}$$

50. We know that $g = \frac{GM}{R^2}$ or $M = \frac{gR^2}{G}$

Here, $g = 1.67 \text{ ms}^{-2}$, $R = 1.74 \times 10^6 \text{ m}$ and

$$G = 6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$$

Therefore we have

$$M = \frac{1.67 \times (1.74 \times 10^6)^2}{6.67 \times 10^{-11}} = 7.6 \times 10^{22} \text{ kg}$$

51. The universal law of gravitation was able to explain successfully

- i. The force that binds us to the earth.
- ii. The gravitational force of earth which keeps the atmosphere close to earth.
- iii. The motion of the moon around the earth.
- iv. The gravitational pull of the sun which keep planets revolving around the sun.
- v. The tides formed by the rising falling of water level in the ocean are due to the gravitational force of moon and the sun.

52. Here, Acceleration due to gravity on moon's surface, $g = 1.67 \text{ ms}^{-2}$, Radius of moon, $R = 1.74 \times 10^6 \text{ m}$ and $G = 6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$

We know that $g = \frac{GM}{R^2}$ or $M = \frac{gR^2}{G}$

$$\therefore M = \frac{1.67 \times (1.74 \times 10^6)^2}{6.67 \times 10^{-11}} = 7.6 \times 10^{22} \text{ kg}$$

Therefore, mass of the moon = $7.6 \times 10^{22} \text{ kg}$

53. A weight of an object is directly proportional to the mass of the earth (M) and inversely proportional to the square of the radius of the earth (R), i.e.,

$$\text{A weight of object} \propto \frac{M}{R^2}$$

$$\text{Original weight, } W_o = mg = mG \frac{M}{R^2} \dots\dots\dots(1)$$

According to question, hypothetically, Mass of earth, $M' = 4M$ and Radius of earth, $R' = \frac{R}{2}$.

$$\text{Now, New weight an object} = W_n = mg' = mG \frac{M'}{R'^2} \dots\dots\dots(2)$$

Now, put the values of M' and R' in equation (2), we get.

$$\text{Weight of an object, } W_n = mG \frac{4M}{\left(\frac{R}{2}\right)^2} = (16mG) \frac{M}{R^2} = 16 \times W_o$$

The weight will become 16 times.

54. i. A bucket of water feels heavier when taken out of water because when immersed in water, an upward force, i.e. buoyant force acts on it which is equal to the weight of water displaced by the bucket.
- ii. The buoyant force on a lead object is lesser than the buoyant force on the iron object because lead has greater density, so it displaces a lesser amount of water consequently lesser amount of buoyant force acts on it.
55. The moon moves around earth in an orbit and this circling around the Earth is a kind of projectile. It has just the right speed with right momentum in the horizontal direction with respect to the acceleration towards earth. When the gravitational force of the earth pulls the moon towards the Earth, the surface of the Earth also curves away from the moon. Thus, the moon continues falling forever, but never reaches the Earth. Instead, it starts circling around the Earth at a constant speed.
56. For both the stones, Their initial velocity, $u = 0$
- Acceleration in downward direction = g

Using equation, $S = ut + \frac{1}{2}at^2$

Now, $h = ut + \frac{1}{2}gt^2$

$$\Rightarrow h = 0 + \frac{1}{2}gt^2$$

$$\Rightarrow h = \frac{1}{2}gt^2$$

$$\Rightarrow t = \sqrt{\frac{2h}{g}}$$

Since time period depends upon the height and acceleration due to gravity. Here, both stones fall from same height and planet. Therefore, Both stones will take the same time to reach the ground.

57. i. Since it is known that a body can sink in water only when its weight is greater than the upthrust act on it by the water. But in this case, the upthrust act on the bottle is greater than its weight, that's why it bounces back on the water surface.
ii. A bucket of water weighs less inside the well water, it is because when the bucket immersed in the water fully, upthrust act on it by water which reduces its actual weight.

58. Yes, the apple attracts the earth with same force as the force with which the earth attracts an apple. Because of the large mass of the earth the acceleration produced in it is negligible, which goes unnoticed. Thus, the Earth does not seem to move towards the apple.

59. We know that $F = \frac{GMm}{R^2}$ as a weight of a body is the force with which a body is attracted towards the earth,

$\therefore W = mg = \frac{GMm}{R^2}$, M = mass of earth, m = mass of body, R = radius of earth, G = gravitational constant.

According to question, If the radius of the earth (R') becomes twice of its original radius(R),

Then, New weight, $W' = \frac{GMm}{R'^2} = \frac{GMm}{4R^2} = \frac{W}{4}$

i.e., the weight will be reduced to one-fourth of the original weight of body.

60. According to the equation of motion under gravity,

$$v^2 - u^2 = 2gs$$

where u is the initial velocity of the ball,

v is the final velocity of the ball

s is the height achieved by the ball

g is the acceleration due to gravity

At maximum height, the final velocity of a ball is zero i.e $v=0$

i. $v = u + gt$

$$0 = 49 + (-9.8) \times t$$

$$9.8t = 49$$

$$t = 49/9.8 = 5 \text{ s}$$

$$h = ut + \frac{1}{2}gt^2 = 49 \times 5 + \frac{1}{2} \times 9.8 \times 25$$

$$= 245 - 122.5 = 122.5$$

ii. total time taken to return = $5 + 5 = 10 \text{ s}$

61. Initial speed of a ball, $u = 0.5 \text{ m/s}$

Since the ball is thrown upward, so negative Acceleration, $g = -9.8 \text{ m/s}^2$

Let at height ' h ' a ball is at rest, so Final speed of ball at height ' h ' is $v = 0$

i. We know $v^2 - u^2 = 2gh$

$$0 - (0.5)^2 = 2 \times (-9.8) \times h$$

$$\text{or } -0.25 = -19.6 A$$

$$\text{or } h = \frac{0.25}{19.6} = 0.0127 \text{ m}$$

$$\Rightarrow h = 1.27 \text{ cm}$$

ii. Let, ' t ' be the time to reach the height ' h '. Now, Putting the values in the formula $v = u + gt$

$$0 = 0.5 - 9.8t$$

$$\text{or } t = \frac{0.5}{9.8} = 0.05 \text{ s}$$

62. $F_{\text{gravitation}} = \frac{G \cdot M_e \times m_o}{r^2}$

Where G is gravitational constant

M_e = the mass of the earth

m_o = the mass of the object

r = the distance between the centre of gravity of the object and the centre of earth.